Lecture 26

PID Controller :

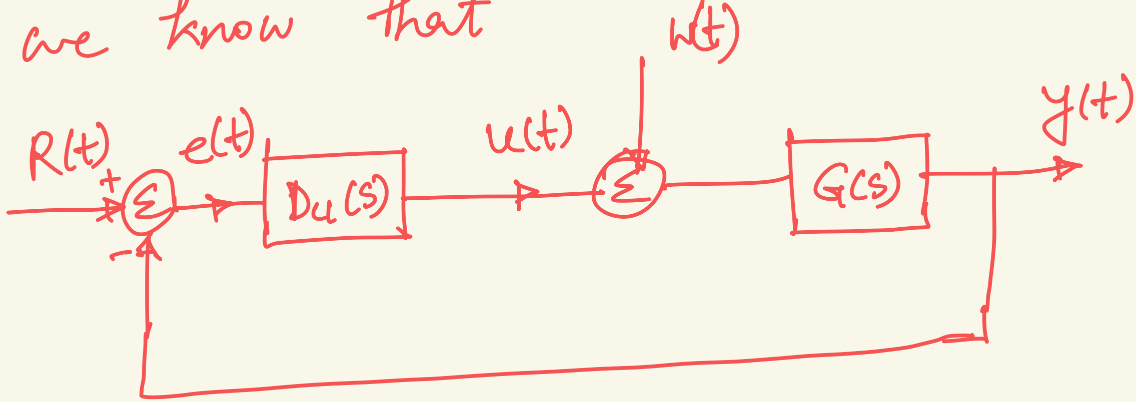
P = Proportional controller

I = Integral controller

D = Derivative controller

Proportional controller :

From the closed loop system we know that



$$u(t) = D_c(s) \cdot e(t)$$

$$\Rightarrow U(s) = k_p \cdot E(s)$$

$$\Rightarrow \frac{U(s)}{E(s)} = k_p$$

k_p is a variable that can be adjusted.

Integral controller:

$$u(t) = k_I \int_{t_0}^{t} e(\tau)\, d\tau$$

$$\Rightarrow \frac{du(t)}{dt} = k_I\, e(t)$$

$$\Rightarrow s\, U(s) = k_I\, E(s)$$

$$\Rightarrow \quad \frac{U(s)}{E(s)} = K_I \cdot \frac{1}{s}$$

Derivative Controller:

$$u(t) = K_D \, \dot{e}(t)$$

$$\Rightarrow \quad \frac{U(s)}{E(s)} = K_D \cdot s$$

For PID controller

$$\frac{U(s)}{E(s)} = D_{cl}(s) = K_P + \frac{K_I}{s} + K_D \cdot s$$

By adjusting K_P, K_I and K_D, appropriate control action can be obtained.